

Face Name Task Behaviour fMRI

Libraries and Functions

```
# Clear environment
rm(list = ls())

# Get Libraries and Functions
source("~/Dropbox (Personal)/TI_paper/ti-paper/Data/Functions.R")
```

Import data and add labels

```
# Import data
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/MRI/FaceNameTask/Behaviour")
data_raw <- read_csv("Behaviour.csv")

# Exclude participant 4 - too much movement in the MRI
data_raw <- subset(data_raw, S_ID!=4)

# Assign appropriate Stim factors
data_raw$StimType <- factor(data_raw$StimType)
levels(data_raw$StimType) <- c("Sham", "TI 1:1", "TI 1:3")

# Assign factors
data_raw$StimType <- factor(data_raw$StimType)
data_raw$block <- factor(data_raw$block)
data_raw$S_ID <- factor(data_raw$S_ID)
```

Check data for outliers - Accuracy level

```
g_data <- ddply(data_raw, c("S_ID"), summarise,
  N = sum(!is.na(trial)),
  mean_Acc_prct = (sum(Accuracy, na.rm=TRUE)/N)*100
)

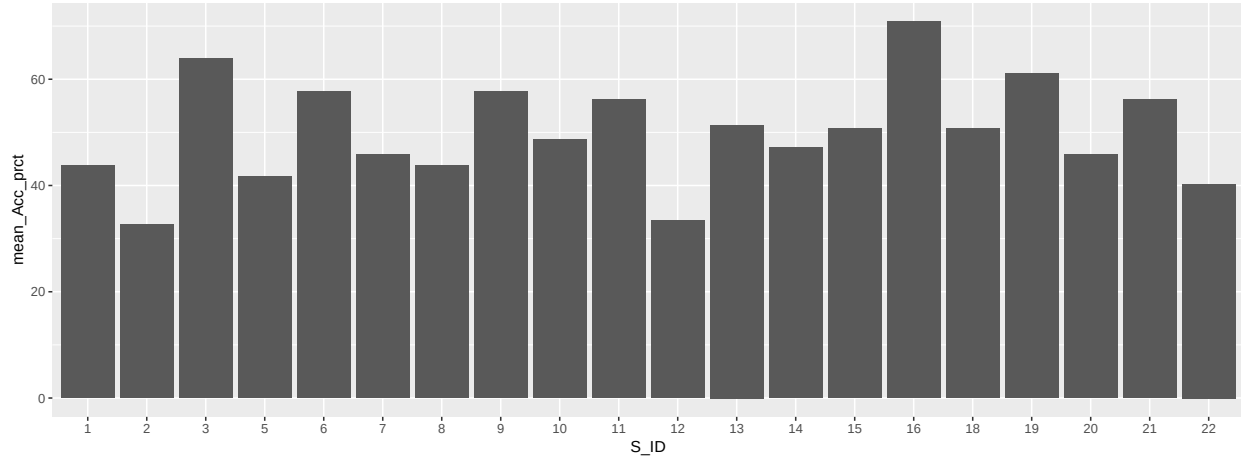
pander(g_data)
```

S_ID	N	mean_Acc_prct
1	144	43.75
2	144	32.64

S_ID	N	mean_Acc_prct
3	144	63.89
5	144	41.67
6	144	57.64
7	144	45.83
8	144	43.75
9	144	57.64
10	144	48.61
11	144	56.25
12	144	33.33
13	144	51.39
14	144	47.22
15	144	50.69
16	144	70.83
18	144	50.69
19	144	61.11
20	144	45.83
21	144	56.25
22	144	40.28

All subjects have mean accuracy above 20% (chance level)

```
ggplot(g_data, aes(x=S_ID, y=mean_Acc_prct)) +
  geom_bar(position=position_dodge(), stat="identity")
```



```
g_data %>% summarise(mean = mean(mean_Acc_prct),
  sd = sd(mean_Acc_prct), min = min(mean_Acc_prct), max = max(mean_Acc_prct))
```

```
##      mean      sd      min      max
## 1 49.96528 9.770336 32.63889 70.83333
```

Check response distributions

```
## What is the distribution of responses (correct, foil, distractors, omission)
bdata <- data_raw
bdata$RespType <- ifelse(bdata$Accuracy==1, "Target",
  ifelse(bdata$Foil==1, "Foil",
    ifelse(bdata$Distractor==1, "Distractor", NA)))

# ----- Checking number of cases (percent) -----

pander(bdata %>%
  group_by(RespType) %>%
  summarise(percent = 100 * n() / nrow(bdata)))
```

RespType	percent
Distractor	16.53
Foil	33.33
Target	49.97
NA	0.1736

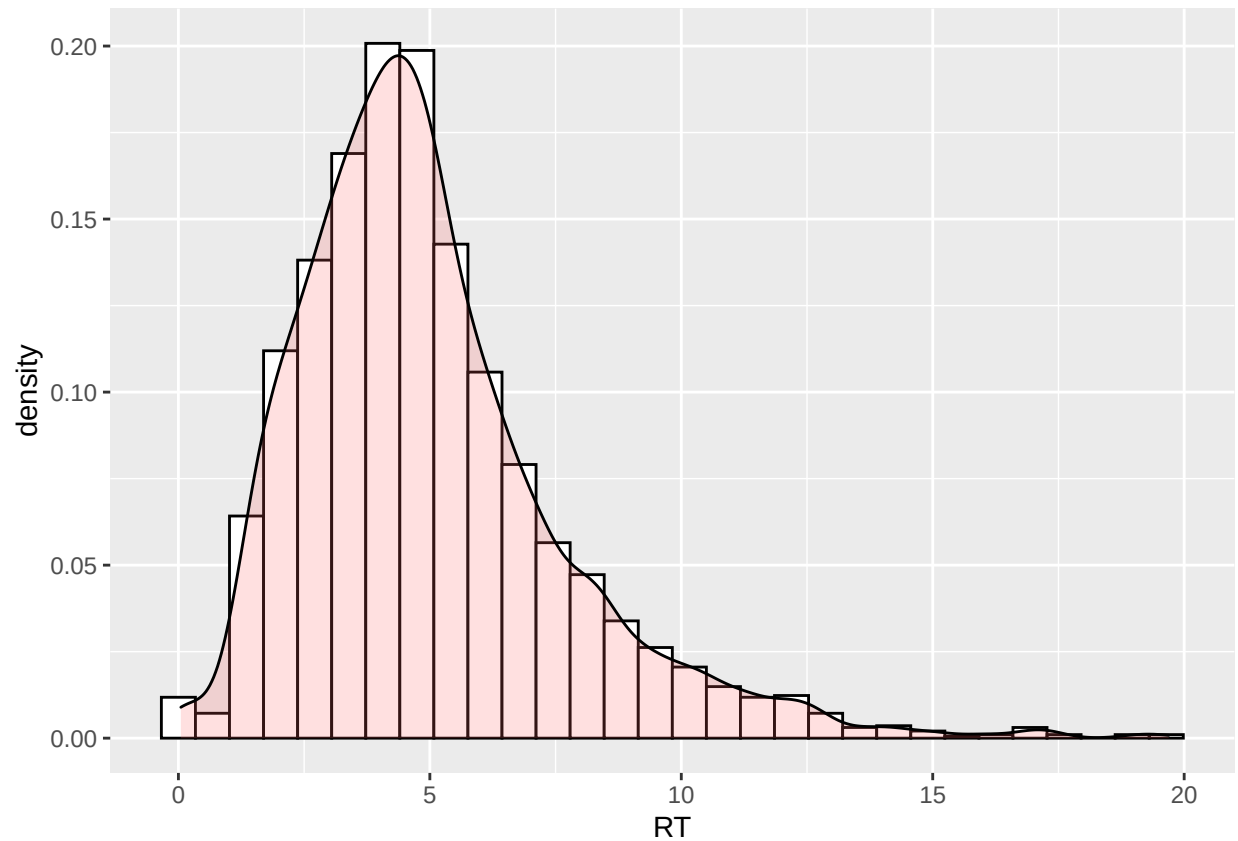
```
# Participants select Correct > Foil > Distractor. They are engaging with the task appropriately. NA =

bdata <- bdata[!is.na(bdata$RespType),] # Getting rid of the few omissions in the dataset
bdata$RespType <- factor(bdata$RespType, levels = c("Target", "Foil", "Distractor"))
```

Check the distribution of RTs and clean the data if needed

```
# ---- Plot histogram of RT -----
ggplot(bdata, aes(x=RT)) +
  geom_histogram(aes(y=..density..), colour="black", fill="white")+
  geom_density(alpha=.2, fill="#FF6666")
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



```
# Typicall histogram distribution, left skew, right tail
```

```
# ----- Check 1 and 99% percentiles and remove those from the data -----
```

```
rts <- bdata$RT
qts <- quantile(rts, probs = c(.01, .99))
qts
```

```
##          1%          99%
## 0.8561875 13.4974937
```

```
bdata <- bdata %>% mutate(RT = replace(RT, RT<qts[1], NA))
bdata <- bdata %>% mutate(RT = replace(RT, RT>qts[2], NA))
bdata <- bdata[!is.na(bdata$RT),]
```

Add additional variables of interest

```
CorrectRT <- ifelse(bdata$Accuracy==1, bdata$RT, NA);
IncorrectRT <- ifelse(bdata$Accuracy==0, bdata$RT, NA);
CorrectConfidence <- ifelse(bdata$Accuracy==1, bdata$Confidence, NA);
IncorrectConfidence <- ifelse(bdata$Accuracy==0, bdata$Confidence, NA);

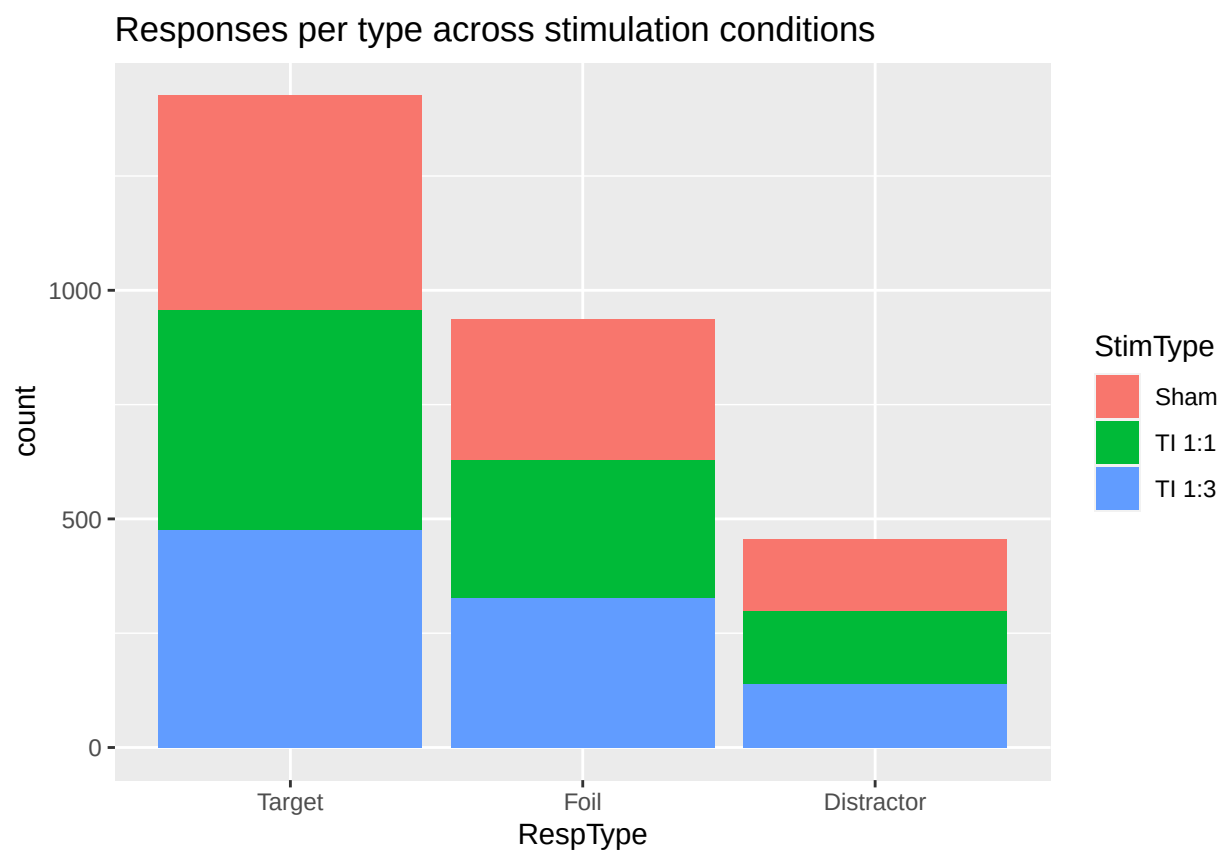
bdata <- cbind(bdata, CorrectRT, IncorrectRT, CorrectConfidence, IncorrectConfidence)
rm(CorrectRT, IncorrectRT, CorrectConfidence, IncorrectConfidence)
```

```
xtabs( ~ RespType + StimType, bdata)
```

Accuracy

```
##           StimType
## RespType  Sham TI 1:1 TI 1:3
##   Target    469  481  476
##   Foil      307  303  326
##   Distractor 156  159  140
```

```
ggplot(bdata, aes(x=RespType)) + geom_bar(aes(fill=StimType),position="stack") +
  ggtitle("Responses per type across stimulation conditions")
```



```
# Does the number of responses per response type vary for different stimulation conditions
bdata$RespType <- relevel(bdata$RespType, ref = "Target")
m.acc <- nnet::multinom(RespType ~ StimType, data = bdata)
```

```
## # weights: 12 (6 variable)
## initial value 3094.790817
## iter 10 value 2830.425815
## iter 10 value 2830.425813
## iter 10 value 2830.425813
## final value 2830.425813
## converged
```

```
summary(m.acc)
```

```
## Call:
## nnet::multinom(formula = RespType ~ StimType, data = bdata)
##
## Coefficients:
##           (Intercept) StimTypeTI 1:1 StimTypeTI 1:3
## Foil           -0.423760   -0.038369485    0.04524322
## Distractor    -1.100753   -0.006205765   -0.12302250
##
## Std. Errors:
##           (Intercept) StimTypeTI 1:1 StimTypeTI 1:3
## Foil           0.07341340    0.1037731    0.1027514
## Distractor    0.09242549    0.1300414    0.1333648
##
## Residual Deviance: 5660.852
## AIC: 5672.852
```

```
exp(coef(m.acc))
```

```
##           (Intercept) StimTypeTI 1:1 StimTypeTI 1:3
## Foil           0.6545809    0.9623573    1.0462823
## Distractor    0.3326205    0.9938135    0.8842438
```

```
# Check the Z-score for the model (wald Z)
```

```
z <- summary(m.acc)$coefficients/summary(m.acc)$standard.errors
z
```

```
##           (Intercept) StimTypeTI 1:1 StimTypeTI 1:3
## Foil           -5.772244   -0.36974406    0.4403174
## Distractor   -11.909628   -0.04772145   -0.9224507
```

```
p <- (1 - pnorm(abs(z), 0, 1))*2
p
```

```
##           (Intercept) StimTypeTI 1:1 StimTypeTI 1:3
## Foil           7.822276e-09    0.7115732    0.6597072
## Distractor    0.000000e+00    0.9619382    0.3562935
```

```
Anova(m.acc)
```

```
## # weights: 6 (2 variable)
## initial value 3094.790817
## final value 2831.640226
## converged
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: RespType
##           LR Chisq Df Pr(>Chisq)
## StimType  2.4288  4    0.6574
```

```
emmeans(m.acc, specs = pairwise ~ StimType|RespType, type = "response")
```

```
## $emmeans
## RespType = Target:
##   StimType   prob      SE df lower.CL upper.CL
##   Sham      0.503 0.0086  6   0.4822   0.524
##   TI 1:1    0.510 0.0158  6   0.4714   0.549
##   TI 1:3    0.505 0.0296  6   0.4330   0.578
##
## RespType = Foil:
##   StimType   prob      SE df lower.CL upper.CL
##   Sham      0.329 0.0207  6   0.2788   0.380
##   TI 1:1    0.321 0.0170  6   0.2797   0.363
##   TI 1:3    0.346 0.0298  6   0.2731   0.419
##
## RespType = Distractor:
##   StimType   prob      SE df lower.CL upper.CL
##   Sham      0.167 0.0176  6   0.1244   0.210
##   TI 1:1    0.169 0.0176  6   0.1255   0.212
##   TI 1:3    0.149 0.0225  6   0.0936   0.204
##
## Confidence level used: 0.95
##
## $contrasts
## RespType = Target:
##   contrast      estimate      SE df t.ratio p.value
##   Sham - TI 1:1  -0.00685 0.0171  6  -0.401  0.9163
##   Sham - TI 1:3  -0.00209 0.0286  6  -0.073  0.9971
##   TI 1:1 - TI 1:3  0.00477 0.0357  6   0.134  0.9902
##
## RespType = Foil:
##   contrast      estimate      SE df t.ratio p.value
##   Sham - TI 1:1   0.00808 0.0242  6   0.333  0.9411
##   Sham - TI 1:3  -0.01667 0.0268  6  -0.623  0.8135
##   TI 1:1 - TI 1:3 -0.02476 0.0321  6  -0.772  0.7322
##
## RespType = Distractor:
##   contrast      estimate      SE df t.ratio p.value
##   Sham - TI 1:1  -0.00123 0.0152  6  -0.081  0.9964
##   Sham - TI 1:3   0.01876 0.0148  6   1.270  0.4599
##   TI 1:1 - TI 1:3  0.01999 0.0256  6   0.780  0.7278
##
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
# No. there is an effect of foil and distractor, but not of stimulation
```

```
#----- binomial -----
```

```
m1 <- glmer(Accuracy ~ StimType + (1|S_ID) , data=bdata, family = binomial(link = "logit"), control=glm
```

```
m2<- glmer(Accuracy ~ StimType + (1|S_ID) + (1|block), data=bdata, family = binomial(link = "logit"), c
```

```
m3 <- glmer(Accuracy ~ StimType + (1|S_ID) + (1|block/trial), data=bdata, family = binomial(link = "log
```

```
anova(m1, m2, m3)
```

```
## Data: bdata
## Models:
## m1: Accuracy ~ StimType + (1 | S_ID)
## m2: Accuracy ~ StimType + (1 | S_ID) + (1 | block)
## m3: Accuracy ~ StimType + (1 | S_ID) + (1 | block/trial)
##      npar    AIC      BIC logLik deviance  Chisq Df Pr(>Chisq)
## m1      4 3860.8 3884.5 -1926.4   3852.8
## m2      5 3853.1 3882.8 -1921.5   3843.1 9.6588  1  0.001884 **
## m3      6 3854.4 3890.0 -1921.2   3842.4 0.7392  1  0.389912
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
lmtest::lrtest(m1, m2, m3)
```

```
## Likelihood ratio test
##
## Model 1: Accuracy ~ StimType + (1 | S_ID)
## Model 2: Accuracy ~ StimType + (1 | S_ID) + (1 | block)
## Model 3: Accuracy ~ StimType + (1 | S_ID) + (1 | block/trial)
##      #Df LogLik Df  Chisq Pr(>Chisq)
## 1      4 -1926.4
## 2      5 -1921.5  1 9.6588  0.001884 **
## 3      6 -1921.2  1 0.7392  0.389912
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# keeping m2
```

```
Anova(m2)
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: Accuracy
##           Chisq Df Pr(>Chisq)
## StimType 0.0583  2    0.9713
```

```
emmeans(m2, specs = pairwise ~ StimType, type = "response" )
```

```
## $emmeans
## StimType prob      SE df asymp.LCL asymp.UCL
## Sham      0.504 0.0302 Inf    0.445    0.563
## TI 1:1     0.509 0.0301 Inf    0.451    0.568
## TI 1:3     0.507 0.0301 Inf    0.448    0.566
##
## Confidence level used: 0.95
## Intervals are back-transformed from the logit scale
##
## $contrasts
```



```
## contrast      odds.ratio      SE  df z.ratio p.value
## Sham / TI 1:1      0.978 0.0922 Inf -0.240  0.9687
## Sham / TI 1:3      0.987 0.0932 Inf -0.141  0.9891
## TI 1:1 / TI 1:3      1.009 0.0953 Inf  0.099  0.9946
##
## P value adjustment: tukey method for comparing a family of 3 estimates
## Tests are performed on the log odds ratio scale
```

```
summary(m2)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial (logit)
## Formula: Accuracy ~ StimType + (1 | S_ID) + (1 | block)
## Data: bdata
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05))
##
##      AIC      BIC    logLik deviance df.resid
## 3853.1 3882.8 -1921.5 3843.1      2812
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.5963 -0.9596  0.6490  0.9649  1.4718
##
## Random effects:
## Groups Name      Variance Std.Dev.
## S_ID (Intercept) 0.1329  0.3645
## block (Intercept) 0.0310  0.1761
## Number of obs: 2817, groups: S_ID, 20; block, 9
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    0.01503   0.12063   0.125   0.901
## StimTypeTI 1:1  0.02264   0.09427   0.240   0.810
## StimTypeTI 1:3  0.01329   0.09441   0.141   0.888
##
## Correlation of Fixed Effects:
##              (Intr) STTI1:1
## StmTypTI1:1 -0.392
## StmTypTI1:3 -0.393  0.500
```

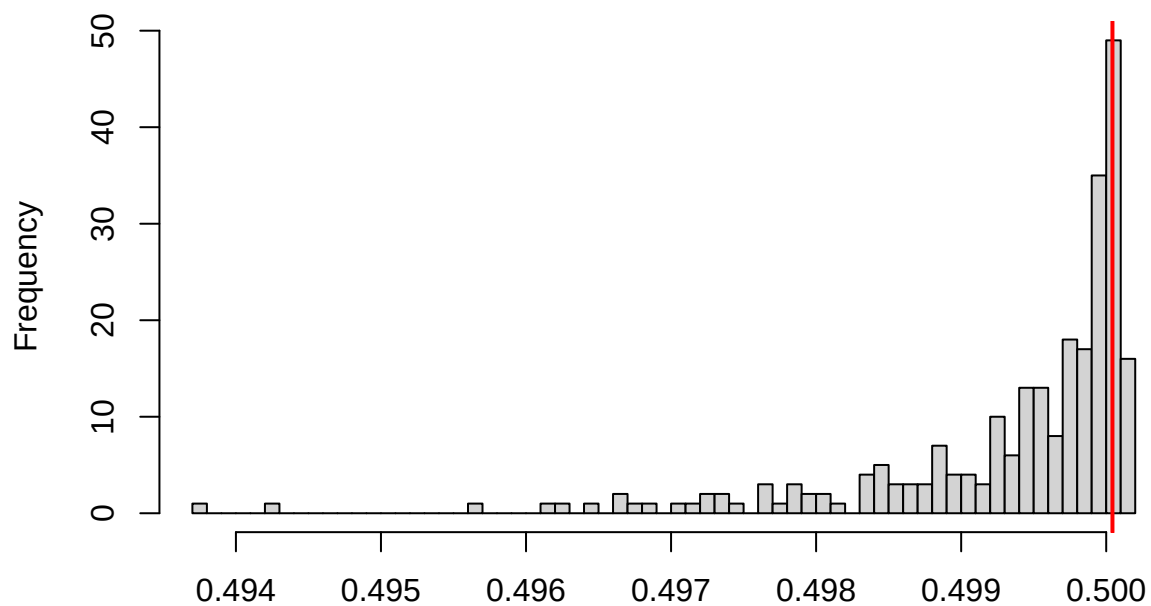
```
library(DHARMA)
```

```
## Warning: package 'DHARMA' was built under R version 3.6.2
```

```
## This is DHARMA 0.3.3.0. For overview type '?DHARMA'. For recent changes, type news(package = 'DHARMA')
```

```
testDispersion(m2)
```

**DHARMA nonparametric dispersion test via sd of
residuals fitted vs. simulated**

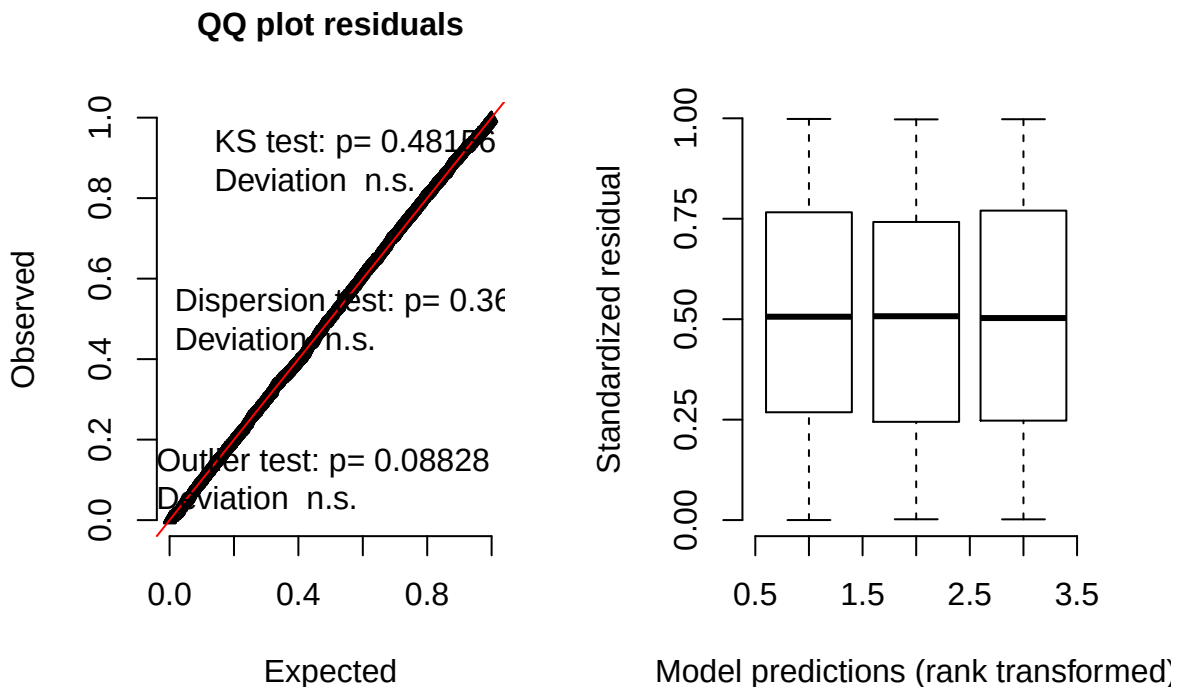


Simulated values, red line = fitted model. p-value (two.sided) = 0.368

```
##  
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.  
## simulated  
##  
## data: simulationOutput  
## ratioObsSim = 1.0014, p-value = 0.368  
## alternative hypothesis: two.sided
```

```
simulationOutput <- simulateResiduals(fittedModel = m2, plot = T)
```

DHARMA residual diagnostics



```
# Checking results on averaged data for completeness
# ----- lmer - group data -----

g_bdata <- ddply(bdata, c("S_ID", "StimType", "RespType"), summarise,
  N = sum(!is.na(trial)),
  meanRespType = N/48, #because there are 48 trials per stimType per participant
  meanRespTypePerc = meanRespType*100,
  medianRT = median(RT, na.rm=TRUE),
  mean_Conf = mean(Confidence, na.rm=TRUE)
)

g_bdata <- g_bdata[!is.na(g_bdata$RespType),] # Getting rid of the few omissions in the dataset
g_bdata$RespType <- factor(g_bdata$RespType, levels = c("Target", "Foil", "Distractor"))

Resptype.acc <- lmer(meanRespType ~ StimType*RespType + (1|S_ID), data = g_bdata, REML = TRUE)

## boundary (singular) fit: see help('isSingular')

Anova(Resptype.acc, test="F")

## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: meanRespType
##              F Df Df.res Pr(>F)
## StimType    0.0270  2   152 0.9734
```

```

## RespType          171.9659  2    152 <2e-16 ***
## StimType:RespType  0.3057  4    152 0.8738
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(Resptype.acc, ddf ="Kenward-Roger")

## Linear mixed model fit by REML. t-tests use Kenward-Roger's method [
## lmerModLmerTest]
## Formula: meanRespType ~ StimType * RespType + (1 | S_ID)
## Data: g_bdata
##
## REML criterion at convergence: -276.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.93953 -0.58581 -0.06277  0.55705  2.62570
##
## Random effects:
## Groups Name Variance Std.Dev.
## S_ID (Intercept) 0.000000 0.00000
## Residual 0.009916 0.09958
## Number of obs: 180, groups: S_ID, 20
##
## Fixed effects:
##
## Estimate Std. Error df t value
## (Intercept) 0.488542 0.022266 171.000000 21.941
## StimTypeTI 1:1 0.012500 0.031489 152.000000 0.397
## StimTypeTI 1:3 0.007292 0.031489 152.000000 0.232
## RespTypeFoil -0.168750 0.031489 152.000000 -5.359
## RespTypeDistractor -0.326042 0.031489 152.000000 -10.354
## StimTypeTI 1:1:RespTypeFoil -0.016667 0.044532 152.000000 -0.374
## StimTypeTI 1:3:RespTypeFoil 0.012500 0.044532 152.000000 0.281
## StimTypeTI 1:1:RespTypeDistractor -0.009375 0.044532 152.000000 -0.211
## StimTypeTI 1:3:RespTypeDistractor -0.023958 0.044532 152.000000 -0.538
## Pr(>|t|)
## (Intercept) < 2e-16 ***
## StimTypeTI 1:1 0.692
## StimTypeTI 1:3 0.817
## RespTypeFoil 3.05e-07 ***
## RespTypeDistractor < 2e-16 ***
## StimTypeTI 1:1:RespTypeFoil 0.709
## StimTypeTI 1:3:RespTypeFoil 0.779
## StimTypeTI 1:1:RespTypeDistractor 0.834
## StimTypeTI 1:3:RespTypeDistractor 0.591
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) StTTI1:1 StTTI1:3 RspTyF RspTyD STTI1:1:RTF STTI1:3:RTF
## StmTypTI1:1 -0.707
## StmTypTI1:3 -0.707 0.500
## RespTypeFol -0.707 0.500 0.500
## RspTypDstrc -0.707 0.500 0.500 0.500

```

```
## STTI1:1:RTF  0.500 -0.707  -0.354  -0.707 -0.354
## STTI1:3:RTF  0.500 -0.354  -0.707  -0.707 -0.354  0.500
## STTI1:1:RTD  0.500 -0.707  -0.354  -0.354 -0.707  0.500      0.250
## STTI1:3:RTD  0.500 -0.354  -0.707  -0.354 -0.707  0.250      0.500
##          STTI1:1:RTD
## StmTypTI1:1
## StmTypTI1:3
## RespTypeFol
## RspTypDstrc
## STTI1:1:RTF
## STTI1:3:RTF
## STTI1:1:RTD
## STTI1:3:RTD  0.500
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

Means to report

```
g_bdata %>%
  group_by(RespType) %>%
  summarise_at(vars(meanRespType), list(mean = mean, std = sd, SE = se))
```

```
## # A tibble: 3 x 4
##   RespType    mean    std    SE
##   <fct>      <dbl> <dbl> <dbl>
## 1 Target      0.495 0.118 0.0152
## 2 Foil         0.325 0.0918 0.0118
## 3 Distractor  0.158 0.0813 0.0105
```

```
g_bdata %>%
  group_by(StimType, RespType) %>%
  summarise_at(vars(meanRespType), list(mean = mean, std = sd))
```

```
## # A tibble: 9 x 4
## # Groups:   StimType [3]
##   StimType RespType    mean    std
##   <fct>    <fct>      <dbl> <dbl>
## 1 Sham    Target      0.489 0.129
## 2 Sham    Foil         0.320 0.104
## 3 Sham    Distractor  0.162 0.0992
## 4 TI 1:1  Target      0.501 0.129
## 5 TI 1:1  Foil         0.316 0.0982
## 6 TI 1:1  Distractor  0.166 0.0772
## 7 TI 1:3  Target      0.496 0.100
## 8 TI 1:3  Foil         0.340 0.0731
## 9 TI 1:3  Distractor  0.146 0.0669
```

Means per participant

```
g_bdata2 <- ddply(bdata, c("S_ID", "StimType", "Accuracy"), summarise,
  N = sum(!is.na(trial)),
  meanRespType = N/48, #because there are 48 trials per stimType per participant
  meanRespTypePerc = meanRespType*100,
```

```

        medianRT = median(RT, na.rm=TRUE),
        mean_Conf = mean(Confidence, na.rm=TRUE)
    )

CorrectTrials <- subset(g_bdata2, Accuracy==1)

CorrectTrial <- CorrectTrials %>%
  tibble::as_tibble() %>%
  group_by(StimType) %>%
  mutate(avg_meanRespTypePerc = mean(meanRespTypePerc)) %>%
  mutate(demeaned_meanRespTypePerc = meanRespTypePerc - avg_meanRespTypePerc)

```

```

# plotting the data
library(ggribes)

```

Reaction Times

```
## Warning: package 'ggribes' was built under R version 3.6.2
```

```
##
```

```
## Attaching package: 'ggribes'
```

```
## The following objects are masked from 'package:tidybayes':
```

```
##
```

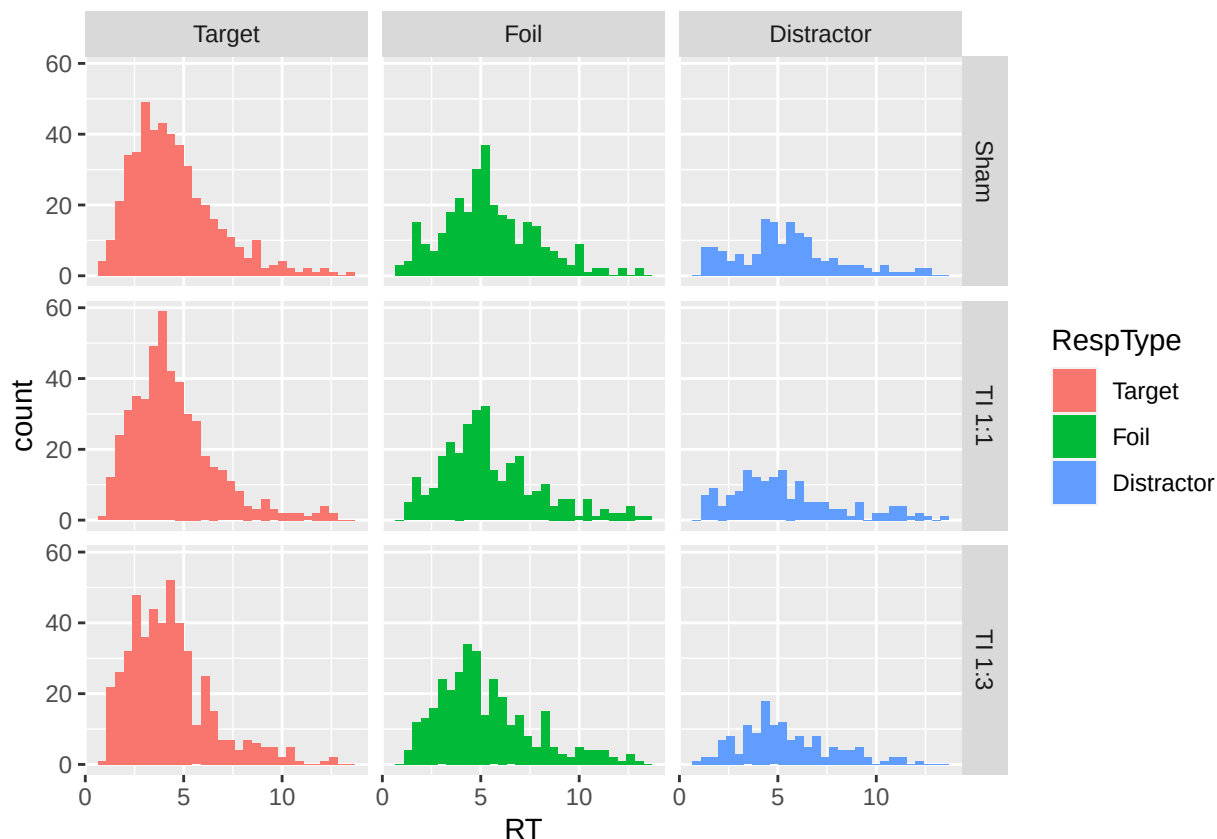
```
##   scale_point_color_continuous, scale_point_color_discrete,
##   scale_point_colour_continuous, scale_point_colour_discrete,
##   scale_point_fill_continuous, scale_point_fill_discrete,
##   scale_point_size_continuous

```

```

ggplot(bdata, aes(x = RT, fill = RespType)) +
  geom_histogram(bins = 30) +
  facet_grid(StimType ~ RespType)

```



```
# Does reaction time per response type vary for different stimulation conditions?

# Using glm without random effects to find starting values for the optimisation
m0 <- glm(RT ~ StimType*RespType, data=bdata, family = inverse.gaussian(link=identity))

m1 <- glmer(RT ~ StimType*RespType + (1|S_ID),
            data = bdata,
            family = inverse.gaussian(link=identity),
            start=list(fixef=coef(m0)),
            control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))

m2 <- glmer(RT ~ StimType*RespType + (1|S_ID) + (1|block),
            data = bdata,
            family = inverse.gaussian(link=identity),
            start=list(fixef=coef(m0)),
            control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))

m3 <- glmer(RT ~ StimType*RespType + (1|S_ID) + (1|block/trial),
            data = bdata,
            family = inverse.gaussian(link=identity),
            start=list(fixef=coef(m0)),
            control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))

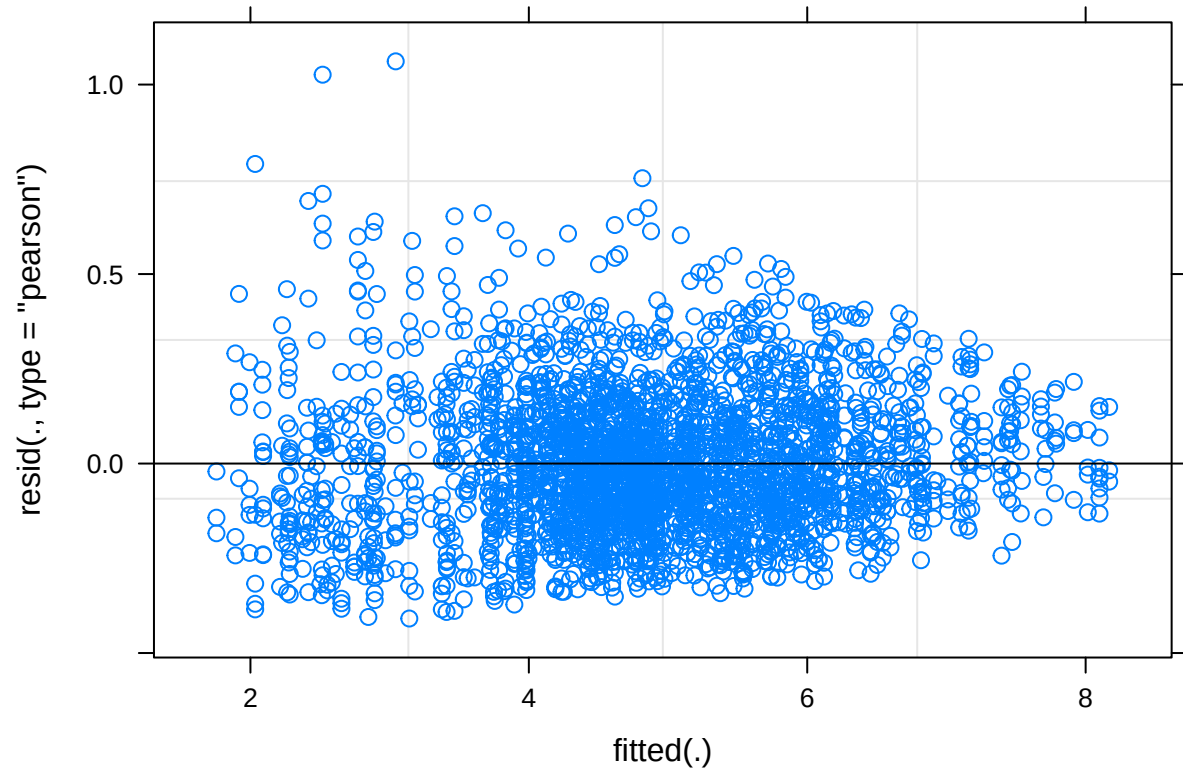
anova(m1, m2, m3)
```

```
## Data: bdata
## Models:
## m1: RT ~ StimType * RespType + (1 | S_ID)
## m2: RT ~ StimType * RespType + (1 | S_ID) + (1 | block)
## m3: RT ~ StimType * RespType + (1 | S_ID) + (1 | block/trial)
##      npar   AIC    BIC logLik deviance   Chisq Df Pr(>Chisq)
## m1     11 11609 11675 -5793.6    11587
## m2     12 11508 11579 -5741.9    11484 103.421   1 < 2.2e-16 ***
## m3     13 11469 11546 -5721.3    11443  41.199   1 1.375e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

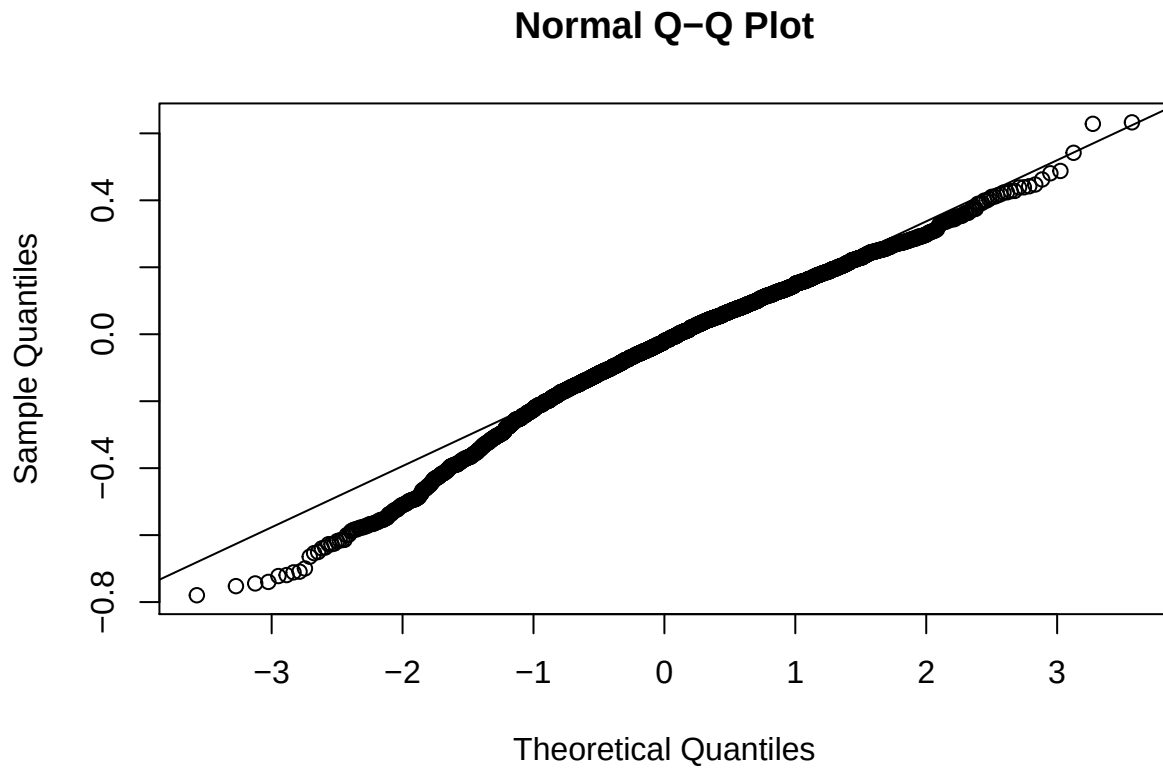
```
lmtest::lrtest(m1, m2, m3)
```

```
## Likelihood ratio test
##
## Model 1: RT ~ StimType * RespType + (1 | S_ID)
## Model 2: RT ~ StimType * RespType + (1 | S_ID) + (1 | block)
## Model 3: RT ~ StimType * RespType + (1 | S_ID) + (1 | block/trial)
##      #Df LogLik Df   Chisq Pr(>Chisq)
## 1     11 -5793.6
## 2     12 -5741.9   1 103.421 < 2.2e-16 ***
## 3     13 -5721.3   1  41.199 1.375e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# keeping m2
plot(m2)
```

```
qqnorm(resid(m2))  
qqline(resid(m2))
```



```
Anova(m2)
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: RT
##               Chisq Df Pr(>Chisq)
## StimType      1.0652  2    0.5871
## RespType     68.2424  2  1.518e-15 ***
## StimType:RespType 6.8863  4    0.1420
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(m2, specs = pairwise ~ StimType|RespType, type = "response" )
```

```
## $emmeans
## RespType = Target:
##   StimType emmean    SE df asymp.LCL asymp.UCL
## Sham      5.24 0.450 Inf     4.36     6.12
## TI 1:1     5.42 0.450 Inf     4.54     6.31
## TI 1:3     5.26 0.448 Inf     4.38     6.14
##
## RespType = Foil:
##   StimType emmean    SE df asymp.LCL asymp.UCL
## Sham      5.86 0.456 Inf     4.97     6.75
## TI 1:1     5.75 0.455 Inf     4.85     6.64
```

```

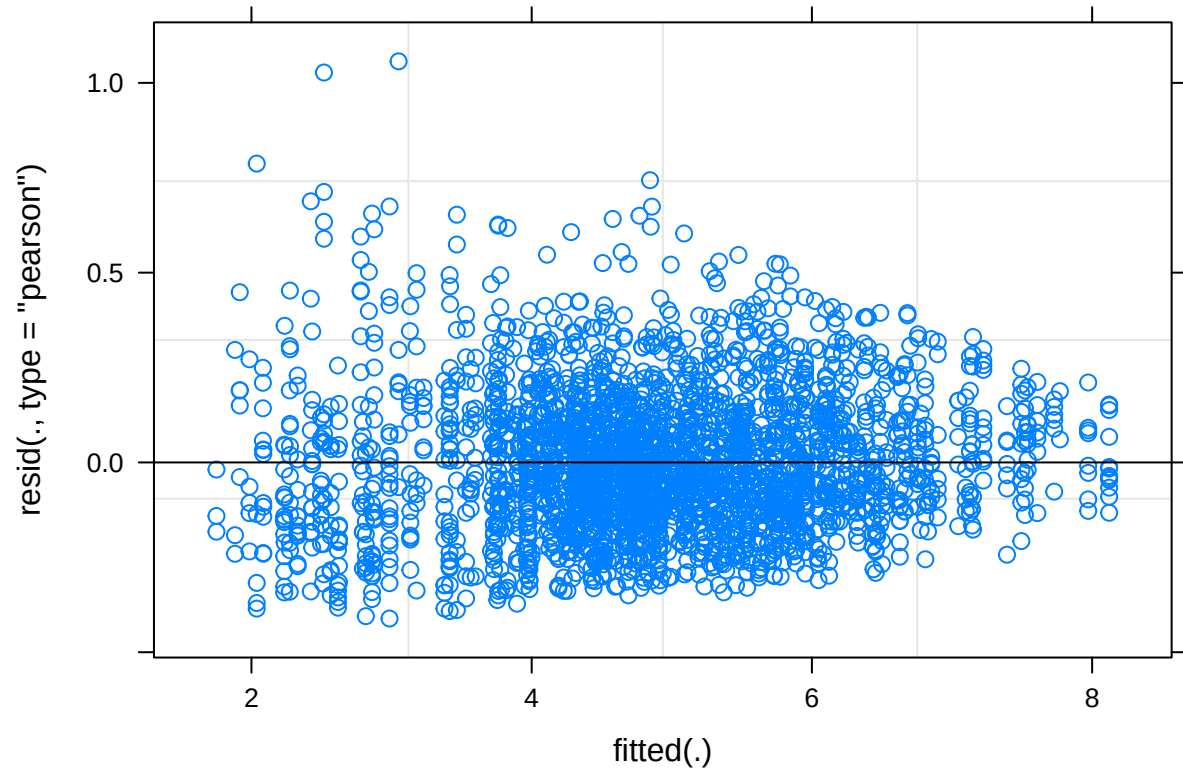
## TI 1:3      5.89 0.456 Inf      4.99      6.78
##
## RespType = Distractor:
## StimType emmean SE df asymp.LCL asymp.UCL
## Sham      5.76 0.462 Inf      4.85      6.67
## TI 1:1     5.82 0.462 Inf      4.91      6.72
## TI 1:3     6.14 0.475 Inf      5.21      7.07
##
## Confidence level used: 0.95
##
## $contrasts
## RespType = Target:
## contrast      estimate      SE df z.ratio p.value
## Sham - TI 1:1   -0.1832 0.1042 Inf -1.757 0.1842
## Sham - TI 1:3   -0.0189 0.0996 Inf -0.190 0.9803
## TI 1:1 - TI 1:3  0.1643 0.1040 Inf  1.579 0.2548
##
## RespType = Foil:
## contrast      estimate      SE df z.ratio p.value
## Sham - TI 1:1    0.1156 0.1539 Inf  0.751 0.7329
## Sham - TI 1:3   -0.0246 0.1578 Inf -0.156 0.9867
## TI 1:1 - TI 1:3 -0.1402 0.1550 Inf -0.904 0.6375
##
## RespType = Distractor:
## contrast      estimate      SE df z.ratio p.value
## Sham - TI 1:1   -0.0544 0.1804 Inf -0.301 0.9511
## Sham - TI 1:3   -0.3792 0.2158 Inf -1.757 0.1841
## TI 1:1 - TI 1:3 -0.3249 0.2153 Inf -1.509 0.2867
##
## P value adjustment: tukey method for comparing a family of 3 estimates

# -- Does reaction time for correct vs incorrect vary for different stimulation conditions
m0 <- glm(RT ~ StimType*as.factor(Accuracy), data=bdata, family = inverse.gaussian(link=identity))

m2 <- glmer(RT ~ StimType*as.factor(Accuracy) + (1|S_ID) + (1|block),
  data = bdata,
  family = inverse.gaussian(link=identity),
  start=list(fixef=coef(m0)),
  control=glmerControl(optimizer="bobyqa", optCtrl = list(maxfun=2e5), nAGQ0initStep=FALSE))

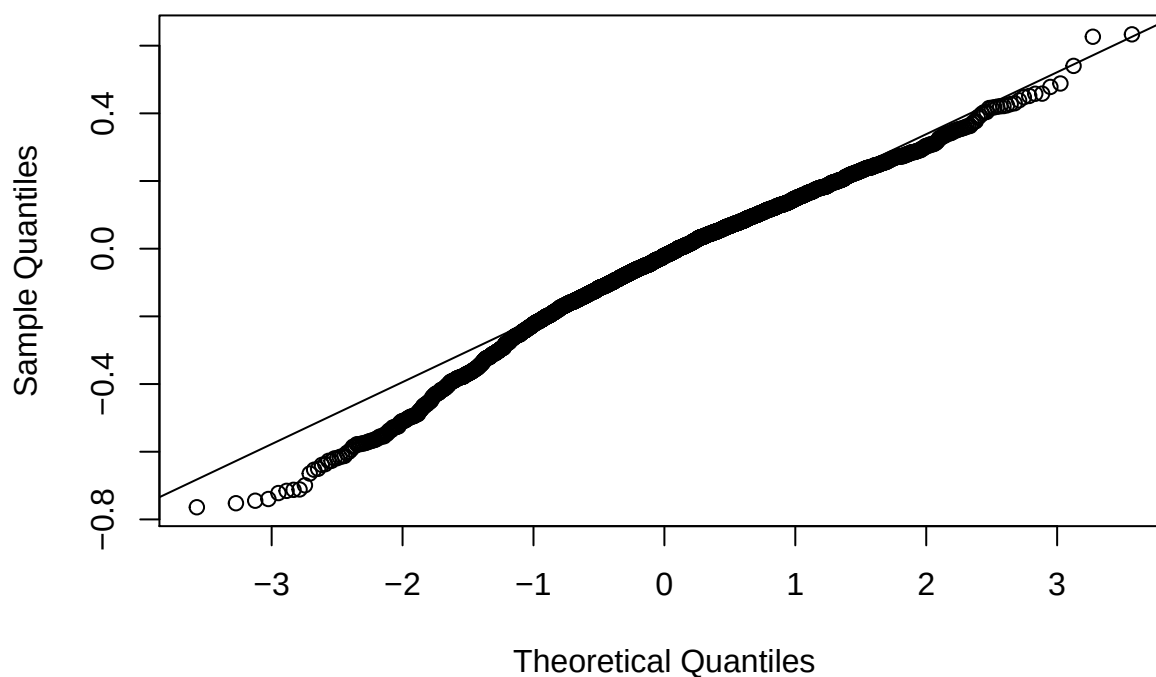
plot(m2)

```



```
qqnorm(resid(m2))  
qqline(resid(m2))
```

Normal Q-Q Plot



```
Anova(m2)
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: RT
##
##               Chisq Df Pr(>Chisq)
## StimType       1.0700  2   0.58567
## as.factor(Accuracy) 69.2122  1   < 2e-16 ***
## StimType:as.factor(Accuracy) 5.1867  2   0.07477 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(m2, specs = pairwise ~ StimType|Accuracy, type = "response" )
```

```
## $emmeans
## Accuracy = 0:
##   StimType emmean    SE df asymp.LCL asymp.UCL
##   Sham      5.82 0.450 Inf      4.94      6.70
##   TI 1:1     5.77 0.450 Inf      4.89      6.66
##   TI 1:3     5.97 0.451 Inf      5.08      6.85
##
## Accuracy = 1:
##   StimType emmean    SE df asymp.LCL asymp.UCL
##   Sham      5.24 0.449 Inf      4.36      6.12
##   TI 1:1     5.42 0.449 Inf      4.54      6.30
```

```
## TI 1:3      5.26 0.448 Inf      4.38      6.14
##
## Confidence level used: 0.95
##
## $contrasts
## Accuracy = 0:
## contrast      estimate      SE df z.ratio p.value
## Sham - TI 1:1      0.0441 0.1174 Inf  0.376  0.9251
## Sham - TI 1:3     -0.1499 0.1272 Inf -1.178  0.4661
## TI 1:1 - TI 1:3   -0.1940 0.1253 Inf -1.548  0.2687
##
## Accuracy = 1:
## contrast      estimate      SE df z.ratio p.value
## Sham - TI 1:1     -0.1842 0.1042 Inf -1.767  0.1807
## Sham - TI 1:3     -0.0186 0.0997 Inf -0.187  0.9809
## TI 1:1 - TI 1:3    0.1656 0.1040 Inf  1.592  0.2492
##
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
estimated_means <- get_means(m2, "StimType*Accuracy")
pander(estimated_means)
```

StimType	Accuracy	Mean	SE	df	CI_lower	CI_higher
Sham	0	5.819	0.4504	Inf	4.936	6.702
TI 1:1	0	5.775	0.4503	Inf	4.892	6.657
TI 1:3	0	5.969	0.4515	Inf	5.084	6.854
Sham	1	5.239	0.4491	Inf	4.359	6.12
TI 1:1	1	5.423	0.4493	Inf	4.543	6.304
TI 1:3	1	5.258	0.4479	Inf	4.38	6.136

```
pander(bdata %>%
  group_by(Accuracy) %>%
  summarise(mean(RT), sd(RT)))
```

Accuracy	mean(RT)	sd(RT)
0	5.395	2.492
1	4.427	2.143

```
# Checking results on averaged data for completeness
```

```
Resptype.rt <- lmer(medianRT ~ StimType*RespType + (1|S_ID), data = g_bdata, REML = TRUE)
Anova(Resptype.rt, test="F")
```

```
## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: medianRT
##              F Df Df.res    Pr(>F)
## StimType    0.3069  2   152    0.7362
## RespType   36.1140  2   152 1.471e-13 ***
```

```

## StimType:RespType 0.4716 4 152 0.7565
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(Resptype.rt, ddf = "Kenward-Roger")

## Linear mixed model fit by REML. t-tests use Kenward-Roger's method [
## lmerModLmerTest]
## Formula: medianRT ~ StimType * RespType + (1 | S_ID)
## Data: g_bdata
##
## REML criterion at convergence: 541.8
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -2.5702 -0.6208 -0.0493 0.5342 3.2841
##
## Random effects:
## Groups Name Variance Std.Dev.
## S_ID (Intercept) 1.9869 1.4096
## Residual 0.8424 0.9178
## Number of obs: 180, groups: S_ID, 20
##
## Fixed effects:
##
## Estimate Std. Error df t value
## (Intercept) 4.19875 0.37612 34.57883 11.163
## StimTypeTI 1:1 0.15865 0.29025 152.00000 0.547
## StimTypeTI 1:3 -0.12371 0.29025 152.00000 -0.426
## RespTypeFoil 1.14941 0.29025 152.00000 3.960
## RespTypeDistractor 1.40154 0.29025 152.00000 4.829
## StimTypeTI 1:1:RespTypeFoil -0.41879 0.41047 152.00000 -1.020
## StimTypeTI 1:3:RespTypeFoil 0.05123 0.41047 152.00000 0.125
## StimTypeTI 1:1:RespTypeDistractor -0.07210 0.41047 152.00000 -0.176
## StimTypeTI 1:3:RespTypeDistractor -0.02840 0.41047 152.00000 -0.069
##
## Pr(>|t|)
## (Intercept) 5.18e-13 ***
## StimTypeTI 1:1 0.585465
## StimTypeTI 1:3 0.670545
## RespTypeFoil 0.000115 ***
## RespTypeDistractor 3.32e-06 ***
## StimTypeTI 1:1:RespTypeFoil 0.309231
## StimTypeTI 1:3:RespTypeFoil 0.900846
## StimTypeTI 1:1:RespTypeDistractor 0.860795
## StimTypeTI 1:3:RespTypeDistractor 0.944921
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) StTTI1:1 StTTI1:3 RspTyF RspTyD STTI1:1:RTF STTI1:3:RTF
## StmTypTI1:1 -0.386
## StmTypTI1:3 -0.386 0.500
## RespTypeFol -0.386 0.500 0.500
## RspTyDstrc -0.386 0.500 0.500 0.500
## STTI1:1:RTF 0.273 -0.707 -0.354 -0.707 -0.354

```

```
## STTI1:3:RTF  0.273 -0.354  -0.707  -0.707 -0.354  0.500
## STTI1:1:RTD  0.273 -0.707  -0.354  -0.354 -0.707  0.500      0.250
## STTI1:3:RTD  0.273 -0.354  -0.707  -0.354 -0.707  0.250      0.500
##           STTI1:1:RTD
## StmTypTI1:1
## StmTypTI1:3
## RespTypeFol
## RspTypDstrc
## STTI1:1:RTF
## STTI1:3:RTF
## STTI1:1:RTD
## STTI1:3:RTD  0.500
```

```
estimated_means <- get_means(Resptype.rt, "StimType*RespType")
pander(estimated_means)
```

StimType	RespType	Mean	SE	df	CI_lower	CI_higher
Sham	Target	4.199	0.3761	34.58	3.435	4.963
TI 1:1	Target	4.357	0.3761	34.58	3.593	5.121
TI 1:3	Target	4.075	0.3761	34.58	3.311	4.839
Sham	Foil	5.348	0.3761	34.58	4.584	6.112
TI 1:1	Foil	5.088	0.3761	34.58	4.324	5.852
TI 1:3	Foil	5.276	0.3761	34.58	4.512	6.04
Sham	Distractor	5.6	0.3761	34.58	4.836	6.364
TI 1:1	Distractor	5.687	0.3761	34.58	4.923	6.451
TI 1:3	Distractor	5.448	0.3761	34.58	4.684	6.212

```
# Save data
save(g_bdata, bdata, file = "BehaviouralData_clean.RData")
```

```
### Means to report
g_bdata %>%
  group_by(StimType, RespType) %>%
  summarise_at(vars(medianRT), list(mean = mean, std = sd))
```

```
## # A tibble: 9 x 4
## # Groups:   StimType [3]
##   StimType RespType    mean    std
##   <fct>    <fct>    <dbl> <dbl>
## 1 Sham     Target     4.20  1.32
## 2 Sham     Foil       5.35  1.62
## 3 Sham     Distractor  5.60  1.94
## 4 TI 1:1   Target     4.36  1.07
## 5 TI 1:1   Foil       5.09  1.71
## 6 TI 1:1   Distractor  5.69  2.32
## 7 TI 1:3   Target     4.08  0.917
## 8 TI 1:3   Foil       5.28  2.00
## 9 TI 1:3   Distractor  5.45  1.75
```



```
xtabs( ~ Confidence + RespType + StimType , bdata)
```

Confidence

```
## , , StimType = Sham
##
##           RespType
## Confidence Target Foil Distractor
##           0      2      0          0
##           1      91  118          89
##           2     120  102          49
##           3     113   64          15
##           4     143   23           3
##
```

```
## , , StimType = TI 1:1
##
##           RespType
## Confidence Target Foil Distractor
##           0      1      1          0
##           1     114  126          74
##           2     104  108          55
##           3     112   53          24
##           4     150   15           6
##
```

```
## , , StimType = TI 1:3
##
##           RespType
## Confidence Target Foil Distractor
##           0      0      1          0
##           1      72  116          60
##           2     132  113          53
##           3     105   75          22
##           4     167   21           5
##
```

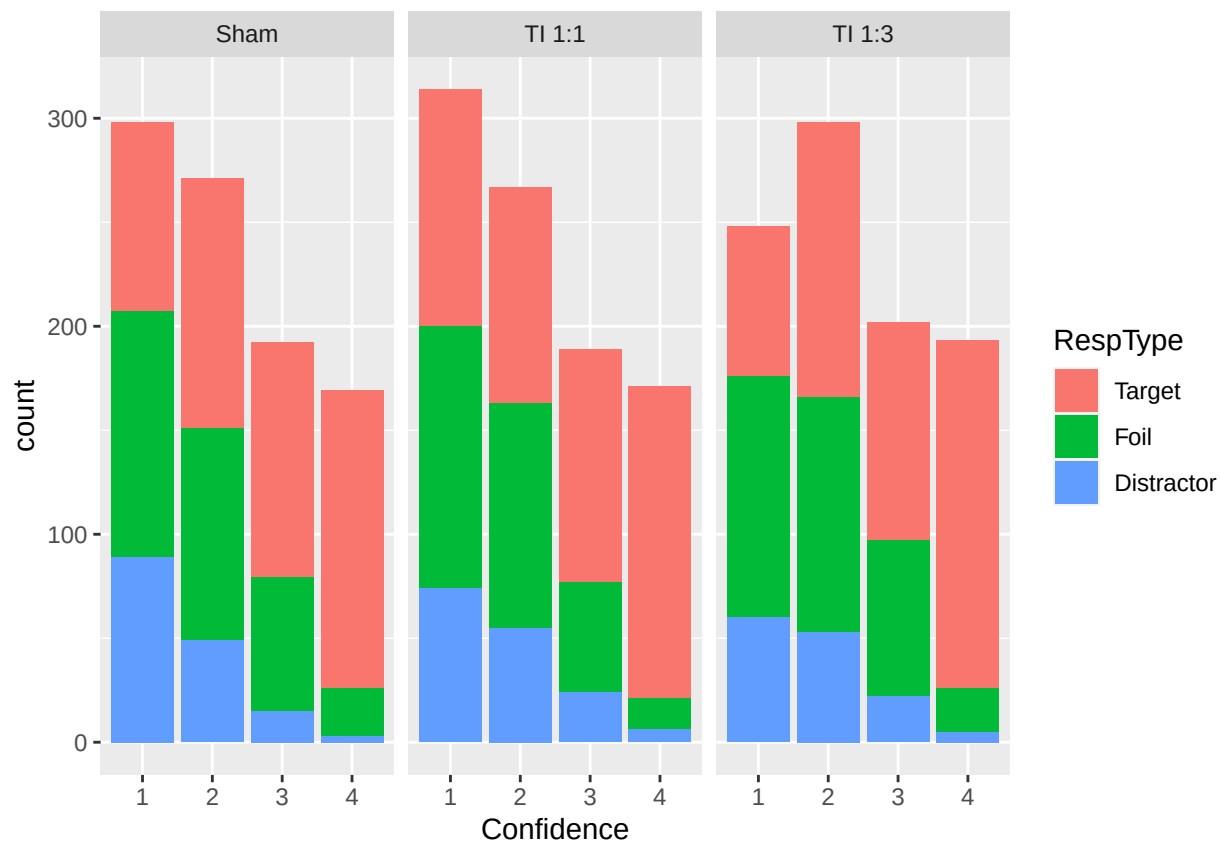
```
# Removing confidence = 0 > no response or technical error
sum(bdata$Confidence=="0")
```

```
## [1] 5
```

```
Confidence_omission_Prct <- (sum(data_raw$Confidence=="0")/length(data_raw$Confidence))*100
```

```
bdata <- bdata %>% mutate(Confidence = replace(Confidence, Confidence=="0", NA))
bdata <- bdata[!is.na(bdata$Confidence),]
bdata$Confidence <- as.factor(bdata$Confidence)
```

```
ggplot(data = bdata) +
  geom_bar(mapping = aes(x = Confidence, fill = RespType)) + facet_wrap(~StimType)
```



```
# Does the accuracy per response type vary for different stimulation conditions
library(ordinal)
```

```
##
## Attaching package: 'ordinal'

## The following objects are masked from 'package:brms':
##
##   ranef, VarCorr

## The following objects are masked from 'package:lme4':
##
##   ranef, VarCorr

## The following object is masked from 'package:dplyr':
##
##   slice
```

```
library(ggeffects)
```

```
## Warning: package 'ggeffects' was built under R version 3.6.2
```

```
m1 = clmm(Confidence ~ StimType*RespType + (1|S_ID), data = bdata, Hess = TRUE)
m2 = clmm(Confidence ~ StimType*RespType + (1|S_ID) + (1|block), data = bdata, Hess = TRUE)
anova(m1,m2)
```

```
## Likelihood ratio tests of cumulative link models:
##
##      formula:                                link: threshold:
## m1 Confidence ~ StimType * RespType + (1 | S_ID)      logit flexible
## m2 Confidence ~ StimType * RespType + (1 | S_ID) + (1 | block) logit flexible
##
##      no.par      AIC  logLik LR.stat df Pr(>Chisq)
## m1          12 7109.6 -3542.8
## m2          13 7087.3 -3530.6  24.282  1  8.322e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
library(RVAideMemoire)
```

```
## Warning: package 'RVAideMemoire' was built under R version 3.6.2
```

```
## *** Package RVAideMemoire v 0.9-79 ***
```

```
##
## Attaching package: 'RVAideMemoire'
```

```
## The following object is masked _by_ '.GlobalEnv':
##
##      se
```

```
## The following objects are masked from 'package:ppcor':
##
##      pcor, pcor.test
```

```
## The following object is masked from 'package:lme4':
##
##      dummy
```

```
Anova(m2)
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: Confidence
##              LR Chisq Df Pr(>Chisq)
## StimType      10.43  2  0.005433 **
## RespType     420.54  2 < 2.2e-16 ***
## StimType:RespType  5.39  4  0.249476
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(m2, specs = pairwise ~ StimType|RespType, type = "response" )
```

```
## $emmeans
## RespType = Target:
##   StimType emmean      SE df asymp.LCL asymp.UCL
##   Sham      0.284 0.166 Inf   -0.0409    0.609
##   TI 1:1     0.213 0.166 Inf   -0.1130    0.538
##   TI 1:3     0.498 0.166 Inf    0.1731    0.822
##
## RespType = Foil:
##   StimType emmean      SE df asymp.LCL asymp.UCL
##   Sham     -0.925 0.176 Inf   -1.2703   -0.580
##   TI 1:1    -1.092 0.178 Inf   -1.4410   -0.744
##   TI 1:3    -0.816 0.174 Inf   -1.1567   -0.476
##
## RespType = Distractor:
##   StimType emmean      SE df asymp.LCL asymp.UCL
##   Sham     -1.820 0.214 Inf   -2.2392   -1.400
##   TI 1:1    -1.413 0.207 Inf   -1.8190   -1.008
##   TI 1:3    -1.287 0.212 Inf   -1.7020   -0.872
##
## Confidence level used: 0.95
##
## $contrasts
## RespType = Target:
##   contrast      estimate      SE df z.ratio p.value
##   Sham - TI 1:1      0.0713 0.122 Inf  0.583  0.8294
##   Sham - TI 1:3     -0.2136 0.122 Inf -1.758  0.1839
##   TI 1:1 - TI 1:3    -0.2849 0.122 Inf -2.331  0.0516
##
## RespType = Foil:
##   contrast      estimate      SE df z.ratio p.value
##   Sham - TI 1:1      0.1672 0.149 Inf  1.125  0.4987
##   Sham - TI 1:3     -0.1087 0.144 Inf -0.755  0.7307
##   TI 1:1 - TI 1:3    -0.2760 0.146 Inf -1.891  0.1413
##
## RespType = Distractor:
##   contrast      estimate      SE df z.ratio p.value
##   Sham - TI 1:1     -0.4065 0.216 Inf -1.879  0.1447
##   Sham - TI 1:3     -0.5325 0.222 Inf -2.396  0.0437
##   TI 1:1 - TI 1:3    -0.1261 0.215 Inf -0.585  0.8281
##
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
m3 = clmm(Confidence ~ StimType + (1|S_ID) + (1|block), data = bdata, Hess = TRUE)
Anova(m3)
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: Confidence
##           LR Chisq Df Pr(>Chisq)
## StimType  10.052  2  0.006566 **
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(m3, specs = pairwise ~ StimType, type = "response" )
```

```
## $emmeans
##   StimType emmean      SE df asymp.LCL asymp.UCL
##   Sham      -0.442 0.152 Inf    -0.740   -0.1447
##   TI 1:1     -0.476 0.152 Inf    -0.774   -0.1778
##   TI 1:3     -0.229 0.151 Inf    -0.526    0.0676
##
## Confidence level used: 0.95
##
## $contrasts
##   contrast      estimate      SE df z.ratio p.value
##   Sham - TI 1:1      0.0335 0.0846 Inf   0.397  0.9170
##   Sham - TI 1:3     -0.2131 0.0841 Inf  -2.533  0.0304
##   TI 1:1 - TI 1:3    -0.2466 0.0847 Inf  -2.912  0.0100
##
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
# Also looking at the average confidence levels for completeness
```

```
Resptype.conf <- lmer(mean_Conf ~ StimType*RespType + (1|S_ID), data = g_bdata, REML = TRUE)
Anova(Resptype.conf, test="F")
```

```
## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: mean_Conf
##              F Df Df.res Pr(>F)
## StimType      1.2000  2    152 0.3040
## RespType     91.6232  2    152 <2e-16 ***
## StimType:RespType 0.5821  4    152 0.6761
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(Resptype.conf, ddf="Kenward-Roger")
```

```
## Linear mixed model fit by REML. t-tests use Kenward-Roger's method [
## lmerModLmerTest]
## Formula: mean_Conf ~ StimType * RespType + (1 | S_ID)
##   Data: g_bdata
##
## REML criterion at convergence: 240.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.7999 -0.5260 -0.0768  0.5301  3.1222
##
## Random effects:
##   Groups   Name      Variance Std.Dev.
##   S_ID      (Intercept) 0.0896   0.2993
##   Residual              0.1683   0.4103
```

```

## Number of obs: 180, groups: S_ID, 20
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)    2.626953   0.113558  87.000819  23.133
## StimTypeTI 1:1    0.000537   0.129734 152.000000    0.004
## StimTypeTI 1:3    0.129186   0.129734 152.000000    0.996
## RespTypeFoil    -0.658253   0.129734 152.000000   -5.074
## RespTypeDistractor -1.053573   0.129734 152.000000   -8.121
## StimTypeTI 1:1:RespTypeFoil -0.080717   0.183472 152.000000   -0.440
## StimTypeTI 1:3:RespTypeFoil -0.081185   0.183472 152.000000   -0.442
## StimTypeTI 1:1:RespTypeDistractor  0.183916   0.183472 152.000000    1.002
## StimTypeTI 1:3:RespTypeDistractor  0.033514   0.183472 152.000000    0.183
##
##              Pr(>|t|)
## (Intercept)    < 2e-16 ***
## StimTypeTI 1:1    0.997
## StimTypeTI 1:3    0.321
## RespTypeFoil    1.12e-06 ***
## RespTypeDistractor 1.47e-13 ***
## StimTypeTI 1:1:RespTypeFoil    0.661
## StimTypeTI 1:3:RespTypeFoil    0.659
## StimTypeTI 1:1:RespTypeDistractor 0.318
## StimTypeTI 1:3:RespTypeDistractor 0.855
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) StTTI1:1 StTTI1:3 RspTyF RspTyD STTI1:1:RTF STTI1:3:RTF
## StmTypTI1:1 -0.571
## StmTypTI1:3 -0.571  0.500
## RespTypeFol -0.571  0.500  0.500
## RspTypDstrc -0.571  0.500  0.500  0.500
## STTI1:1:RTF  0.404 -0.707 -0.354 -0.707 -0.354
## STTI1:3:RTF  0.404 -0.354 -0.707 -0.707 -0.354  0.500
## STTI1:1:RTD  0.404 -0.707 -0.354 -0.354 -0.707  0.500  0.250
## STTI1:3:RTD  0.404 -0.354 -0.707 -0.354 -0.707  0.250  0.500
##
##              STTI1:1:RTD
## StmTypTI1:1
## StmTypTI1:3
## RespTypeFol
## RspTypDstrc
## STTI1:1:RTF
## STTI1:3:RTF
## STTI1:1:RTD
## STTI1:3:RTD  0.500

```

```

estimated_means <- get_means(Resptype.conf, "StimType*RespType")
pander(estimated_means)

```

StimType	RespType	Mean	SE	df	CI_lower	CI_higher
Sham	Target	2.627	0.1136	87	2.401	2.853
TI 1:1	Target	2.627	0.1136	87	2.402	2.853

StimType	RespType	Mean	SE	df	CI_lower	CI_higher
TI 1:3	Target	2.756	0.1136	87	2.53	2.982
Sham	Foil	1.969	0.1136	87	1.743	2.194
TI 1:1	Foil	1.889	0.1136	87	1.663	2.114
TI 1:3	Foil	2.017	0.1136	87	1.791	2.242
Sham	Distractor	1.573	0.1136	87	1.348	1.799
TI 1:1	Distractor	1.758	0.1136	87	1.532	1.984
TI 1:3	Distractor	1.736	0.1136	87	1.51	1.962

```
estimated_means <- get_means(Resptype.conf, "StimType")
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```
pander(estimated_means)
```

StimType	Mean	SE	df	CI_lower	CI_higher
Sham	2.056	0.08535	33.88	1.883	2.23
TI 1:1	2.091	0.08535	33.88	1.918	2.265
TI 1:3	2.17	0.08535	33.88	1.996	2.343

```
### Means to report
```

```
g_bdata %>%
  group_by(StimType, RespType) %>%
  summarise_at(vars(mean_Conf), list(mean = mean, std = sd))
```

```
## # A tibble: 9 x 4
## # Groups:   StimType [3]
##   StimType RespType   mean   std
##   <fct>    <fct>    <dbl> <dbl>
## 1 Sham     Target     2.63 0.335
## 2 Sham     Foil       1.97 0.555
## 3 Sham     Distractor 1.57 0.586
## 4 TI 1:1   Target     2.63 0.474
## 5 TI 1:1   Foil       1.89 0.492
## 6 TI 1:1   Distractor 1.76 0.471
## 7 TI 1:3   Target     2.76 0.357
## 8 TI 1:3   Foil       2.02 0.578
## 9 TI 1:3   Distractor 1.74 0.638
```

```
cbPalette <- c("#E69F00", "#D55E00", "#CC79A7", "#999999")
```

```
levels(bdata$StimType)[levels(bdata$StimType)=="Sham"] <- "Sham"
levels(bdata$StimType)[levels(bdata$StimType)=="TI 1:1"] <- "TI 1:1"
levels(bdata$StimType)[levels(bdata$StimType)=="TI 1:3"] <- "TI 1:3"
```

```
library(scales)
```

```
# Plot response type
```

```
bdata$Confidence <- as.numeric(bdata$Confidence)
```

```
g_bdata <- ddply(bdata, c("S_ID", "StimType", "RespType"), summarise,
  N = sum(!is.na(trial)),
  meanRespType = N/48, #because there are 48 trials per stimType per participant
  meanRespTypePerc = meanRespType*100,
  medianRT = median(RT, na.rm=TRUE),
  mean_Conf = mean(Confidence, na.rm=TRUE)
)
```

```
p.RespType <- ggplot(g_bdata, aes(x = RespType, y = meanRespTypePerc, fill = StimType)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width
  scale_y_continuous(name = "Mean Response (%)",
    breaks = seq(0, 100, 20),
    limits=c(0, 100)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    panel.background = element_blank(),
    legend.title=element_blank()) +
  scale_x_discrete(breaks=c("Target","Foil", "Distractor"),
    labels=c("Target","Foil", "Distractor")) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 20),
    legend.text = element_text(size=14),
    legend.key.size = unit(0.5, 'cm')) + ggtitle("") + xlab("") + scale_fill_manual(values = c("Target", "Foil", "Distractor"))
```

```
ggsave(p.RespType ,file="Figure_2C.tiff", width = 12, height = 9, dpi = 300, units = "cm", device = "tiff")
```

```
# alternative
```

```
p.RespType.alt <- ggplot(bdata, aes(x = StimType,
  fill = factor(RespType, levels = c("Target", "Foil", "Distractor")))) +
  geom_bar(position = "fill") +
  scale_y_continuous(breaks = seq(0, 1, .2),
    label = percent) +
  scale_fill_manual(values = cbPalette) +
  labs(y = "Percent", fill = "Response",
    x = "Stimulation",
    title = "") +
  theme_minimal() + theme(axis.text = element_text(size = 12), axis.title = element_text(size = 14),
    legend.text = element_text(size = 12),
```



```

        legend.title = element_blank(),
        axis.title.x = element_blank(),
        axis.text.x.top = element_text(size = 14, face = "bold", color = "black"),
        panel.grid.major = element_blank(), panel.grid.minor = element_blank(), panel
scale_x_discrete(position = "top") + ylab("Response")

### Plot RT

p.RespType.RT <- ggplot(g_bdata, aes(x = RespType, y = medianRT, fill = StimType)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width
scale_y_continuous(name = "Median RT (s)",
  breaks = seq(0, 12, 2),
  limits=c(0, 12)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    panel.background = element_blank(),
    legend.title=element_blank()) +
  scale_x_discrete(breaks=c("Target", "Foil", "Distractor"),
    labels=c("Target", "Foil", "Distractor")) +
    theme(panel.grid.major = element_blank(),
      panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 20),
    legend.text = element_text(size=14),
    legend.key.size = unit(0.5, 'cm')) + ggtitle("") + xlab("") + scale_fill_manual(values = c("

# Plot RT distribution
library(ggribes)

bdata$RespType <- factor(bdata$RespType, levels = c("Distractor", "Foil", "Target"), ordered = TRUE)

p.RT <- ggplot(bdata, aes(x = RT, y = StimType, fill = RespType)) +
  geom_density_ridges(scale = 0.8, alpha = 0.6) +
  theme_ridges(grid = FALSE, center_axis_labels = TRUE) +
  scale_fill_manual(values=c("#E69F00", "#D55E00", "#CC79A7"),
    name="Response",
    breaks=c("Target", "Foil", "Distractor"),
    labels=c("Target", "Foil", "Distractor")) +
  theme(axis.title.y = element_blank(), axis.text.y = element_text(size = 14, face = "bold"),
    legend.title = element_blank()) +
  scale_x_continuous(name = "Reaction Time (s)") + scale_y_discrete(expand = c(0, 0.2))

```

```
# Plot confidence
```

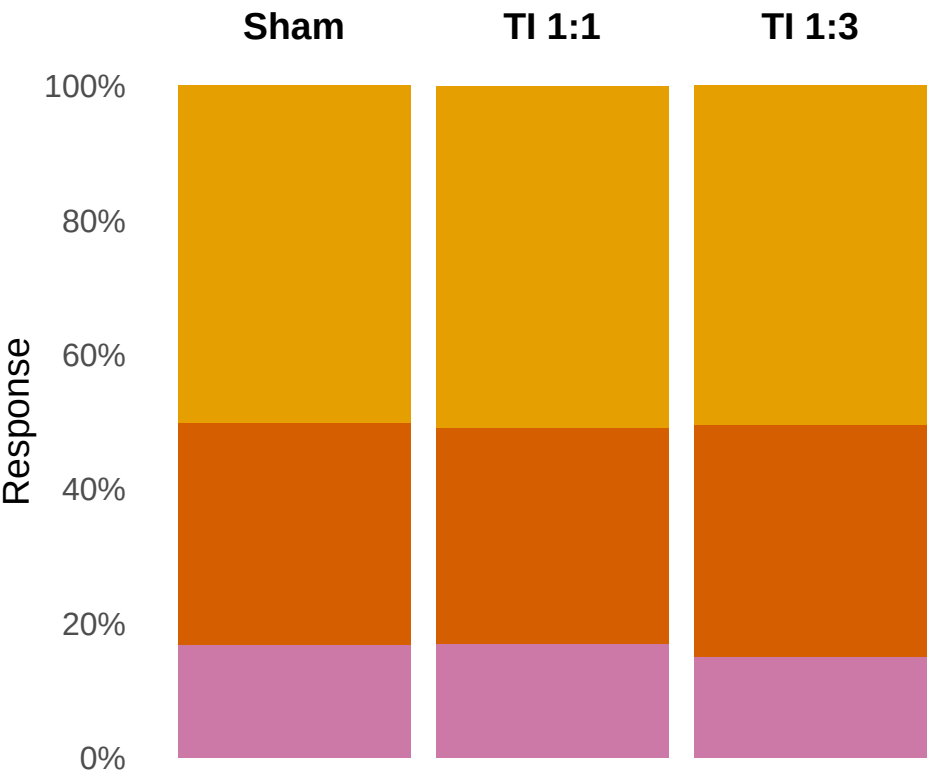
```
level_order <- factor(g_bdata$RespType, levels = c("Target", "Foil", "Distractor"))

p.RespType.Conf <- ggplot(g_bdata, aes(x = level_order, y = mean_Conf, fill = StimType)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width = 0.5) +
  scale_y_continuous(name = "Mean Confidence",
    breaks = seq(0, 4, 1),
    limits = c(0, 4)) +
  theme(plot.title = element_text(color = "black", size = 14, face = "bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key = element_blank(),
    panel.background = element_blank(),
    legend.title = element_blank()) +
  scale_x_discrete(breaks = c("Target", "Foil", "Distractor"),
    labels = c("Target", "Foil", "Distractor")) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 20),
    legend.text = element_text(size = 14),
    legend.key.size = unit(0.5, 'cm')) + ggtitle("") + xlab("") + scale_fill_manual(values = c("Target" = "#f08080", "Foil" = "#4682b4", "Distractor" = "#90ee90"))
```

```
# Alternative
```

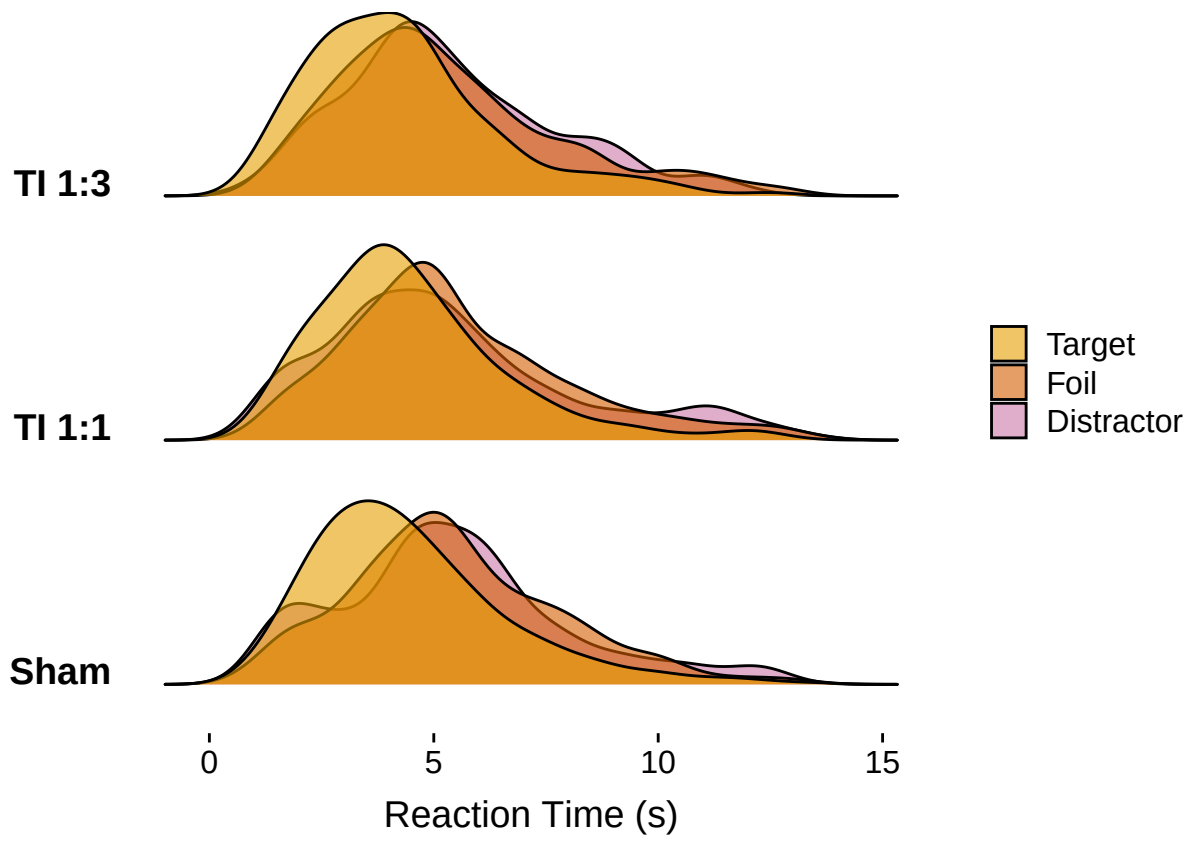
```
p.Confidence <- ggplot(bdata, aes(x = factor(RespType,
  levels = c("Target", "Foil", "Distractor")),
  fill = factor(Confidence, levels = c("4", "3", "2", "1")))) +
  geom_bar(position = "fill") +
  scale_y_continuous(breaks = seq(0, 1, .2),
    label = percent) +
  scale_fill_manual(values = cbPalette) +
  labs(y = "Percent", fill = "Confidence",
    x = "",
    title = "") +
  theme_minimal() + theme(text = element_text(size = 14),
    axis.text = element_text(size = 12),
    axis.title = element_text(size = 14),
    legend.text = element_text(size = 12),
    legend.title = element_blank(),
    axis.title.x = element_blank(),
    axis.text.x.top = element_text(size = 14, face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1),
    panel.grid.major = element_blank(), panel.grid.minor = element_blank(),
    panel.background = element_blank()) + ylab("Confidence") + facet_grid(~StimType)
```

p.RespType.alt

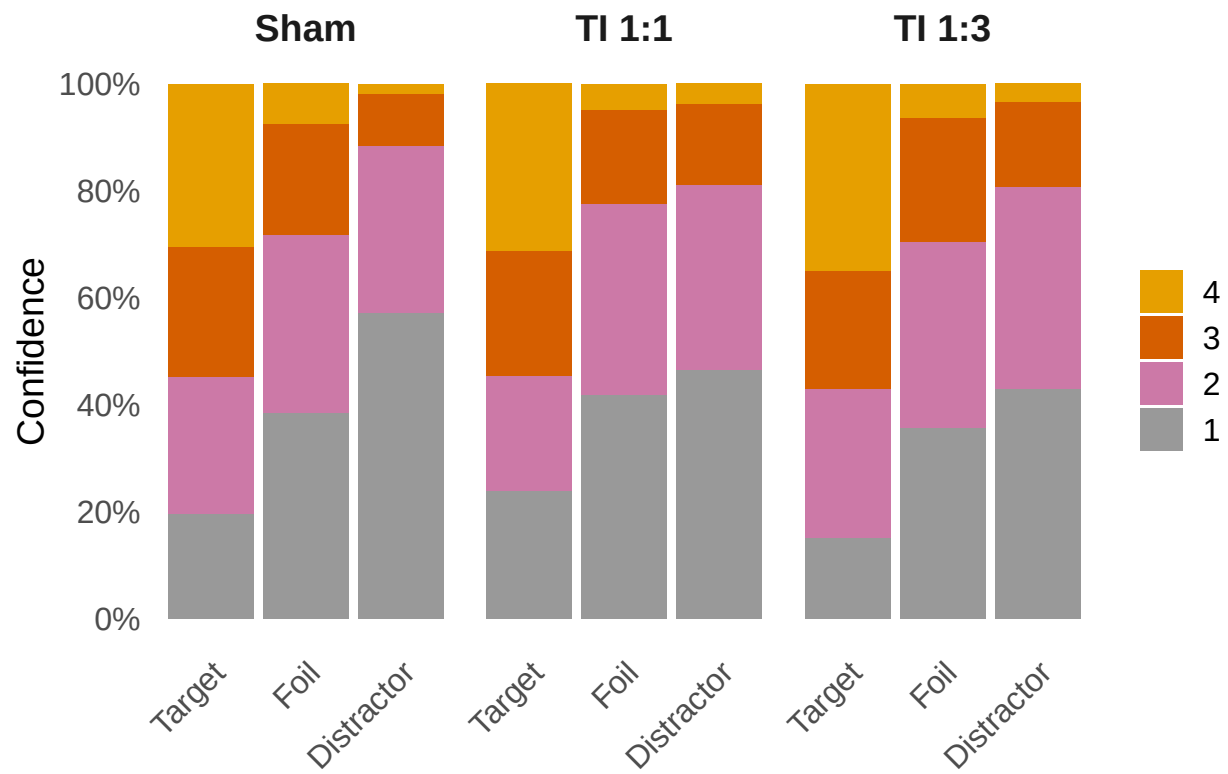


Plots - Figure 1C and Figure S3

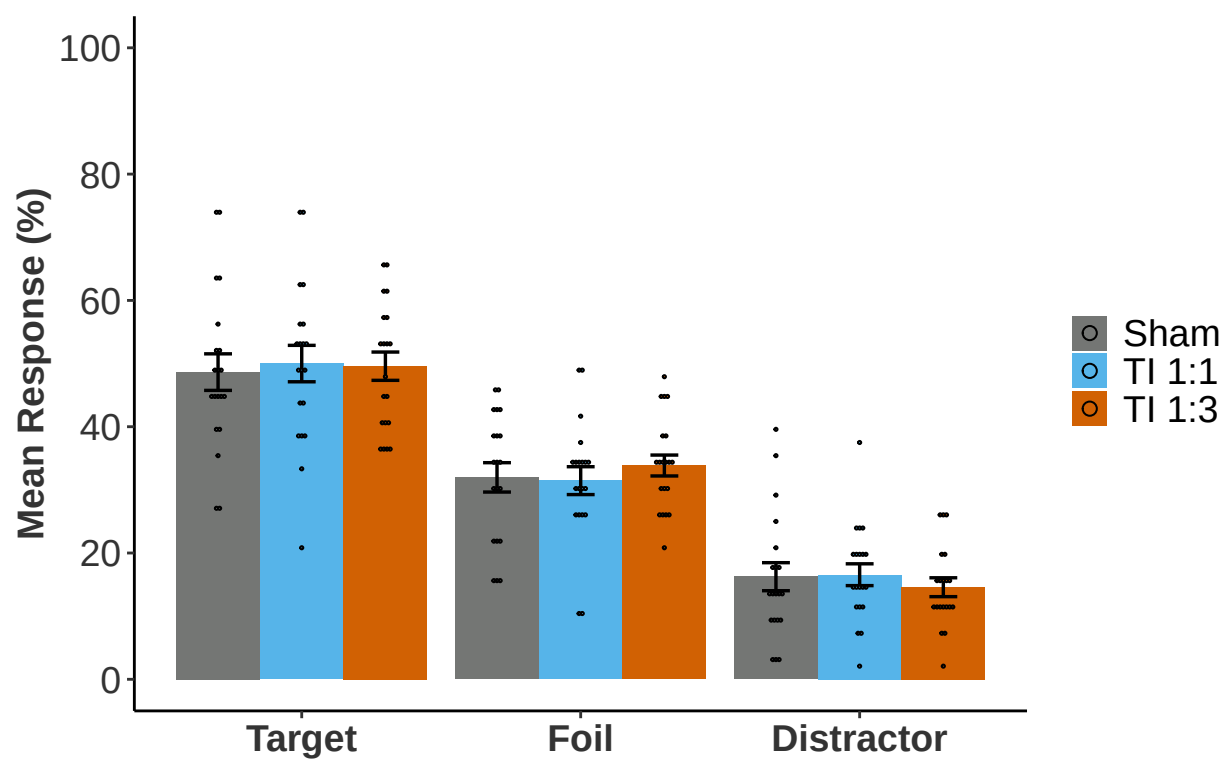
p.RT



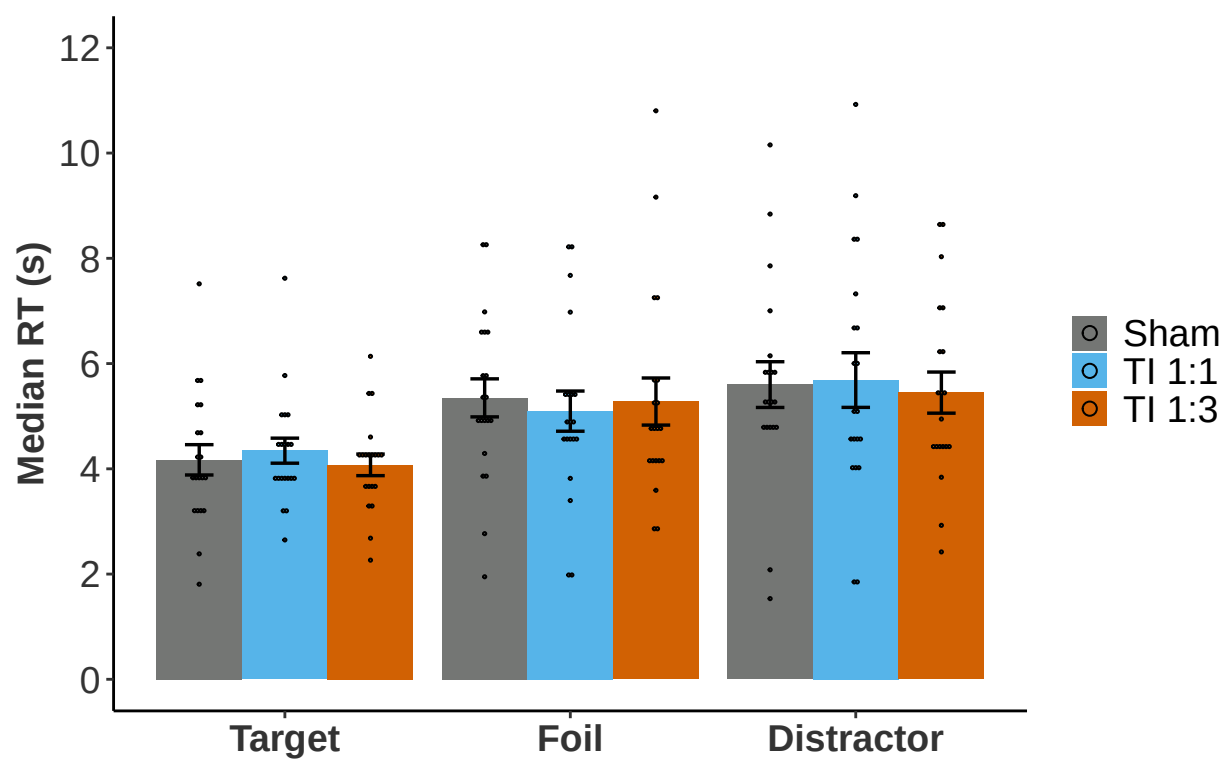
p.Confidence



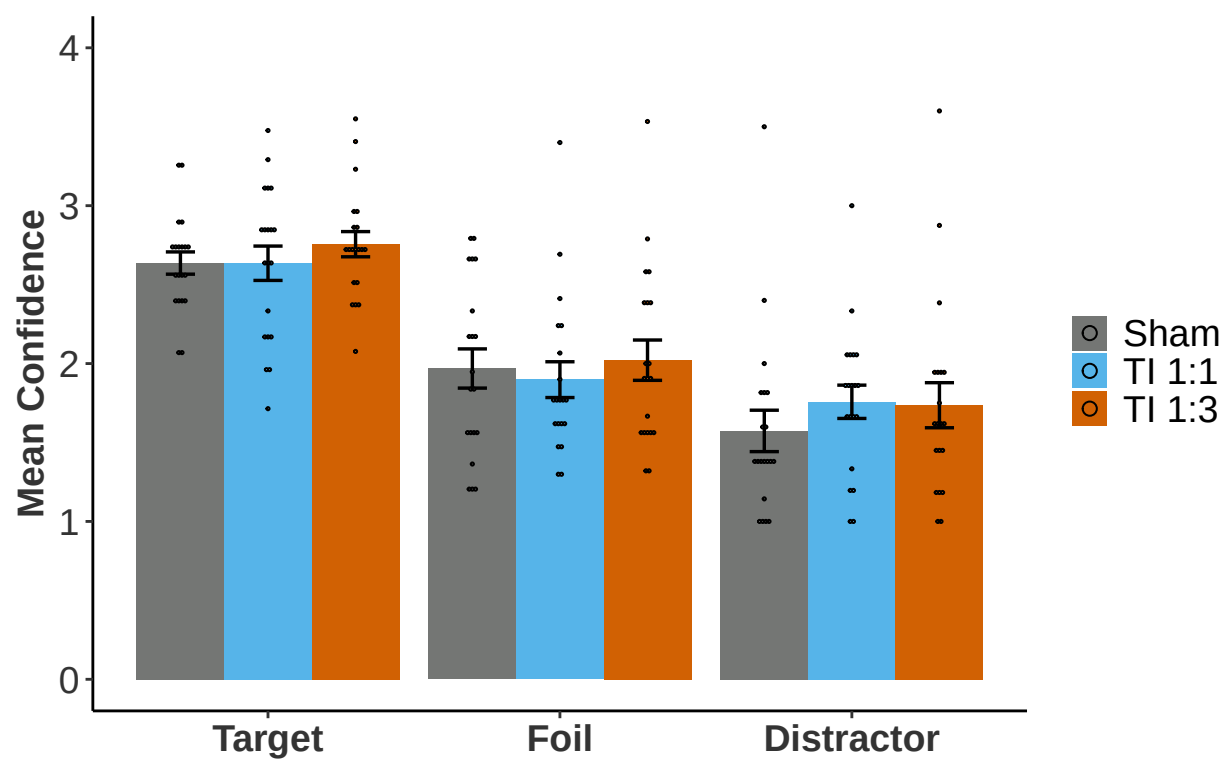
p.RespType



p.RespType.RT

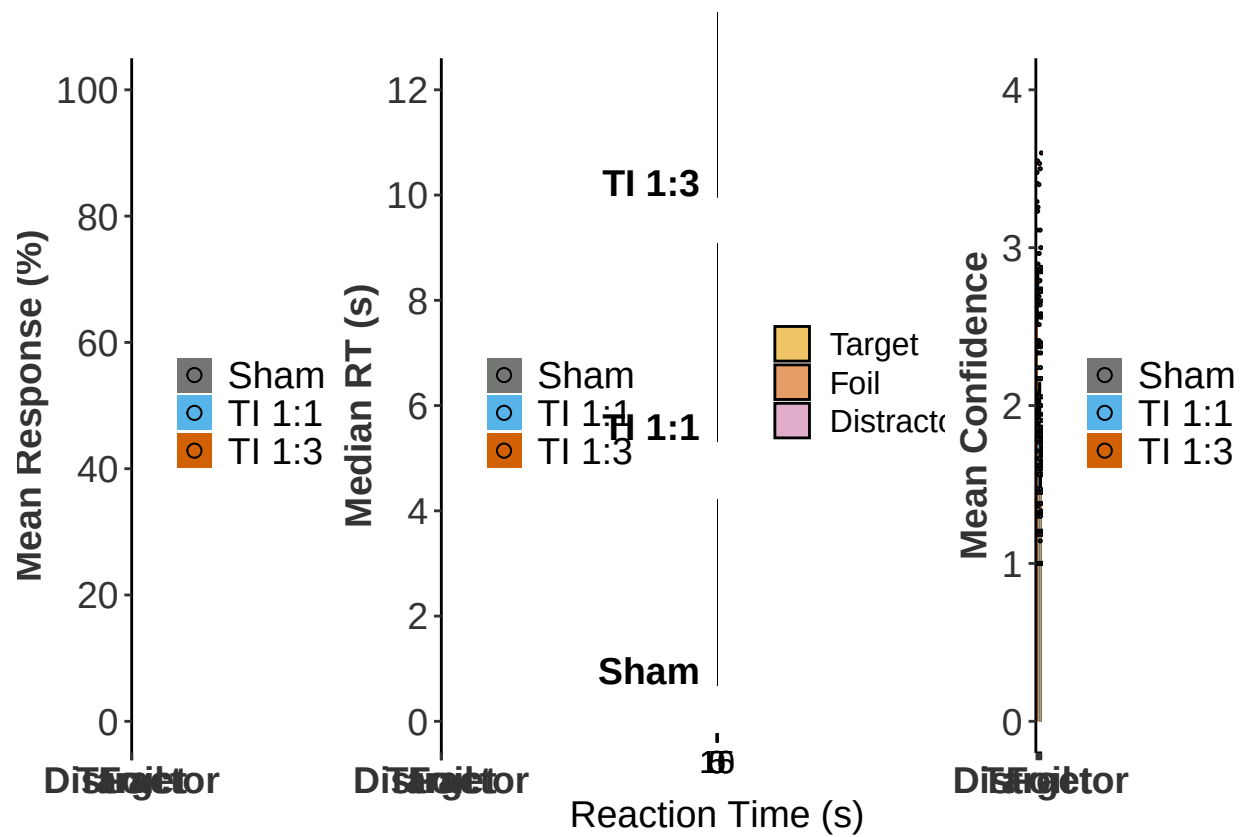


p.RespType.Conf



```
# Combine Plots
```

```
p.Behav <- grid.arrange(p.RespType, p.RespType.RT, p.RT, p.RespType.Conf, nrow = 1, ncol=4, widths = c(
```

```
ggsave(p.Behav, filename = "Figure_S3.tiff", width = 40, height = 10, dpi = 300, units = "cm", device =
```